

PDF

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This covers how to load pdfs into a document format that we can use downstream.

Using PyPDF

Allows for tracking of page numbers as well.

```
from langchain.document_loaders import PagedPDFSplitter

loader = PagedPDFSplitter("example_data/layout-parser-paper.pdf")
pages = loader.load_and_split()
```

```
pages[0]
```

Document(page_content='LayoutParser : A Unified Toolkit for Deep Learning Based Document Image Analysis\nZejiang Shen1(✉), Ruochen Zhang2, Melissa Dell3, Benjamin Charles Germain\nLee4, Jacob Carlson3, and Weining Li5\n1Allen Institute for AI\nshannons@allenai.org\n2Brown University\nruochen.zhang@brown.edu\n3Harvard University\nfmelissadell,jacob.carlson@fas.harvard.edu\n4University of Washington\nnbcgl@cs.washington.edu\n5University of Waterloo\nnw422li@uwaterloo.ca\nAbstract. Recent advances in document image analysis (DIA) have been primarily driven by the application of neural networks. Ideally, research outcomes could be easily deployed in production and extended for further investigation. However, various factors like loosely organized codebases and sophisticated model configurations complicate the easy reuse of important innovations by a wide audience. Though there have been on-going efforts to improve reusability and simplify deep learning (DL) model training pipelines with natural language processing and computer vision', metadata={'source': 'example_data/layout-parser-paper.pdf', 'page': 1})

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represents a major gap in the existing toolkit, as DIA is central to academic research across a wide range of disciplines in the social sciences and humanities. This paper introduces LayoutParser, an open-source library for streamlining the usage of DL in DIA research and applications. The core LayoutParser library comes with a set of simple and intuitive interfaces for applying and customizing DL models for layout detection, character recognition, and many other document processing tasks. To promote extensibility, LayoutParser also incorporates a community platform for sharing both pre-trained models and full document digitization pipelines. We demonstrate that LayoutParser is helpful for both lightweight and large-scale digitization pipelines in real-world use cases. The library is publicly available at <https://layout-parser.github.io>.

Keywords: Document Image Analysis · Deep Learning · Layout Analysis · Character Recognition · Open Source library · Toolkit.

1 Introduction

Deep Learning (DL)-based approaches are the state-of-the-art for a wide range of document image analysis (DIA) tasks including document image classification [11, arXiv:2103.15348v2 [cs.CV] 21 Jun 2021], lookup_str='', metadata={'source': 'example_data/layout-parser-paper.pdf', 'page': '0'}, lookup_index=0)

An advantage of this approach is that documents can be retrieved with page numbers.

```
from langchain.vectorstores import FAISS
from langchain.embeddings.openai import OpenAIEmbeddings

faiss_index = FAISS.from_documents(pages, OpenAIEmbeddings())
docs = faiss_index.similarity_search("How will the community be engaged?", k=2)
for doc in docs:
    print(str(doc.metadata["page"]) + ":", doc.page_content)
```

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Fig. 4: Illustration of (a) the original historical Japanese document with layout detection results and (b) a recreated version of the document image that achieves much better character recognition recall. The reorganization algorithm rearranges the tokens based on the their detected bounding boxes given a maximum allowed height.

4LayoutParser Community Platform

Another focus of LayoutParser is promoting the reusability of layout detection

models and full digitization pipelines. Similar to many existing deep learning

libraries, LayoutParser comes with a community model hub for distributing layout models. End-users can upload their self-trained models to the model hub,

and these models can be loaded into a similar interface available

LayoutParser pre-trained models. For example, the model t



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Beyond DL models, LayoutParser also promotes the sharing of entire document digitization pipelines. For example, sometimes the pipeline requires the combination of multiple DL models to achieve better accuracy. Currently, pipelines are mainly described in academic papers and implementations are often not publicly available. To this end, the LayoutParser community platform also enables the sharing of layout pipelines to promote the discussion and reuse of techniques. For each shared pipeline, it has a dedicated project page, with links to the source code, documentation, and an outline of the approaches. A discussion panel is provided for exchanging ideas. Combined with the core LayoutParser library, users can easily build reusable components based on the shared pipelines and apply them to solve their unique problems.

5 Use Cases

The core objective of LayoutParser is to make it easier to create both large-scale and light-weight document digitization pipelines. Large-scale document processing

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Efficient Data Annotation Customized Model Training
 Model Customization DIA Model Hub DIA Pipeline Sharing Community
 Platform Layout Detection Models Document Images
 The Core LayoutParser Library OCR Module Storage & Visualization
 Layout Data Structure

Fig. 1: The overall architecture of LayoutParser. For an input document image,

the core LayoutParser library provides a set of off-the-shelf tools for layout

detection, OCR, visualization, and storage, backed by a carefully designed layout

data structure. LayoutParser also supports high level customization via efficient

layout annotation and model training functions. These improve model accuracy

on the target samples. The community platform enables the easy sharing of DIA

models and whole digitization pipelines to promote reusability and reproducibility.

A collection of detailed documentation, tutorials and exemplar projects make

LayoutParser easy to learn and use.

AllenNLP [8] and transformers [34] have provided the community with complete

DL-based support for developing and deploying models for general computer vision and natural language processing problems. LayoutParser, on the other

hand, specializes specifically in DIA tasks. LayoutParser is also equipped with a

community platform inspired by established model hubs such as

and TensorFlow Hub [11]. It enables the sharing of pretrained models as



full document processing pipelines that are unique to DIA tasks. There have been a variety of document data collections to facilitate the development of DL models. Some examples include PRImA [3](magazine layouts), PubLayNet [38](academic paper layouts), Table Bank [18](tables in academic papers), Newspaper Navigator Dataset [16,17](newspaper gure layouts) and HJDataset [31](historical Japanese document layouts). A spectrum of models trained on these datasets are currently available in the LayoutParser model zoo to support di erent use cases.

3 The Core LayoutParser Library

At the core of LayoutParser is an o -the-shelf toolkit that streamlines DL-based document image analysis. Five components support a simple interface with comprehensive functionalities: 1) The layout detection models enable using pre-trained or self-trained DL models for layout detection with just four lines of code. 2) The detected layout information is stored in carefully engineered

Using Unstructured

```
from langchain.document_loaders import UnstructuredPDFLoader
```

```
loader = UnstructuredPDFLoader("example_data/layout-parser-paper.pdf")
```

```
data = loader.load()
```

Retain Elements

Under the hood, Unstructured creates different “elements” for different chunks of text. By default we combine those together, but you can easily keep that separation by specifying

```
mode="elements".
```

```
loader = UnstructuredPDFLoader("example_data/layout-parser-paper.pdf",  
mode="elements")
```

```
data = loader.load()
```

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```
data[0]
```

Using PDFMiner

```
from langchain.document_loaders import PDFMinerLoader
```

```
loader = PDFMinerLoader("example_data/layout-parser-paper.pdf")
```

```
data = loader.load()
```

Using PyMuPDF

This is the fastest of the PDF parsing options, and contains detailed metadata about the PDF and its pages, as well as returns one document per page.

```
from langchain.document_loaders import PyMuPDFLoader
```

```
loader = PyMuPDFLoader("example_data/layout-parser-paper.pdf")
```

```
data = loader.load()
```

```
data[0]
```

```
Document(page_content='LayoutParser: A Unified Toolkit for Deep\nLearning Based Document Image Analysis\nZejiang Shen1 (✉), Ruochen Zhang2,\nMelissa Dell3, Benjamin Charles Germain\nLee4, Jacob Carlson3, and\nWeining Li5\n1 Allen Institute for AI\nshannons@allenai.org\n2 Brown University\nruochen_zhang@brown.edu\n3 Harvard University\n{melissadell,jacob_carlson}@fas.harvard.edu\n4 University of Washington\nnbcgl@cs.washington.edu\n5 University of Waterloo\nnw422li@uwaterloo.ca\nAbstract. Recent advances in document image analysis (DIA) have been\nprimarily driven by the application of neural networks. Ideally, research\noutcomes could be easily deployed in production and extended for further\ninvestigation. However, factors like loosely organized codebases\nand sophisticated
```

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reusability and simplify deep learning (DL) model development in disciplines like natural language processing and computer vision, none of them are optimized for challenges in the domain of DIA. This represents a major gap in the existing toolkit, as DIA is central to academic research across a wide range of disciplines in the social sciences and humanities. This paper introduces LayoutParser, an open-source library for streamlining the usage of DL in DIA research and applications. The core LayoutParser library comes with a set of simple and intuitive interfaces for applying and customizing DL models for layout detection, character recognition, and many other document processing tasks. To promote extensibility, LayoutParser also incorporates a community platform for sharing both pre-trained models and full document digitization pipelines. We demonstrate that LayoutParser is helpful for both lightweight and large-scale digitization pipelines in real-world use cases. The library is publicly available at <https://layout-parser.github.io>.
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```
[11, \narXiv:2103.15348v2 [cs.CV] 21 Jun 2021\n', lookup_str='',
metadata={'file_path': 'example_data/layout-parser-paper.pdf',
'page_number': 1, 'total_pages': 16, 'format': 'PDF 1.5', 'title': '',
'author': '', 'subject': '', 'keywords': '', 'creator': 'LaTeX with
hyperref', 'producer': 'pdfTeX-1.40.21', 'creationDate':
'D:20210622012710Z', 'modDate': 'D:20210622012710Z', 'trapped': '',
'encryption': None}, lookup_index=0)
```

Additionally, you can pass along any of the options from the [PyMuPDF documentation](#) as keyword arguments in the `load` call, and it will be pass along to the `get_text()` call.